

1(a)

The sign of β_2 is fairly clear-cut as interest rates rise, stock returns fall, so $\beta_2 < 0$. Higher interest rates imply that T-bill and bond investments are more attractive, and also signal a future slowdown in economic activity.

The sign of β_1 is less clear. While economic growth can be a good thing for the stock market, it can also signal inflation, which tends to depress stock prices.

2 (b)

Source	SS	df	MS	Number of obs = 557		
Model	10737.4639	2	5368.73196	F(2, 554) = 3.33		
Residual	892163.226	554	1610.40293	Prob > F = 0.0364		
				R-squared = 0.0119		
				Adj R-squared = 0.0083		
Total	902900.69	556	1623.9221	Root MSE = 40.13		

rsp500	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
pcip	.0364168	.1293963	0.28	0.778	-.2177506	.2905842
i3	-1.361689	.5407244	-2.52	0.012	-2.423809	-.299568
_cons	18.84306	3.27488	5.75	0.000	12.41036	25.27576

$$\widehat{rsp500}_t = 18.84 + 0.36 pcip_t - 1.36i3_t$$

(3.27) (.129) (0.54)

$$n = 557, R^2 = .012.$$

(c) Only $i3$ is statistically significant with t statistic ≈ -2.52 . The P-value and 95% CI approaches also give the same conclusion.

(d) Clearly, we cannot reject the null hypothesis.

(e) The regression here has nothing directly to say about predicting stock returns because the explanatory variables are dated contemporaneously with $rsp500$. In other words, we do not know $i3_t$ before we know $rsp500_t$. What the regression in part (a) says is that a change in $i3$ is associated with a contemporaneous change in $rsp500$.

(f) Discussed during the tuts

2

(a)

β_1 (**Trend coefficient**):

- This coefficient measures the deterministic time trend in GDP. This variable “t” accounts for long-term growth or decline that is not explained by interest rates.
- If $\beta_1 > 0$, GDP is generally increasing over time.
- If $\beta_1 < 0$ GDP is generally decreasing over time.

β_2 (**Current** interest rate effect)

Given other thing things being equal, when the interest rate increases by 1% at time t, then the GDP will increase by β_2 (\$) on average.

β_3 (**Lagged** interest rate effect) has the similar explanation like we explain β_2 , but it measures the effect of the previous period's interest rate.

(b)

$$Cov(y_t, y_{t-1}) > 0$$

Example: interest rate, GDP, unemployment rate, COE price, stock price, etc.

(c)

$$Cov(y_t, y_{t-1}) < 0$$

Example: temperature (every 12 hours)